88 Other B- and T-Aggressive Lymphomas and Lymphomas Associated with HIV

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88.1 Burkitt Lymphoma

88.1.1 Definition and Epidemiology

BL accounts for around 2% of all adult NHL with a higher incidence in patients with immunodeficiency and in patients who have HIV infection. There is an endemic pediatric subtype in equatorial Africa, which is strongly associated with EBV. The clinical course of BL usually is highly aggressive with a Ki67 expression of nearly 100% requiring prompt institution of therapy.

88.1.2 Diagnosis

A tissue biopsy/cytology sample is mandatory for the diagnosis. The translocation of *MYC* with the immunoglobulin heavy-chain loci (*IgH*), 80% or less frequently, with kappa or lambda light-chains, is the molecular hallmark of BL. Diagnosis requires a combination of morphology, immunophenotype, and genetic analysis.

88.1.3 Prognostic Factors

Several studies have identified prognostic factors for poor outcome. The Burkitt lymphoma International Prognostic Index (BL-IPI) identified four variables (age ≥40 years, performance status ≥2, serum LDH >3 × upper limit of normal, and CNS involvement) with independent prognostic value for PFS and OS (Olszewski et al. 2021).

88.1.4 First Line Treatment

The optimal first-line therapy in BL has not been defined yet. To achieve a fast and stable remission, an intensive regimen combining several compounds is used in most centers. Combinations of rituximab, DOX, alkylators, vincristine, and VP with direct therapy to prevent CNS disease are highly active. Frequently used regimens are the HOVON R-CODOX-M/R-IVAC (RTX, CY, vincristine, DOX, MTX, IFO, Ara-C, VP) (Mead et al. 2002), R-HyperCVAD (RTX, hyperfractionated CY, vincristine, DOX, dexamethasone) (Thomas et al. 1999), and the GMALL-B-ALL-protocol (RTX, MTX, Ara-C, CY, VP, IFO)

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(Hoelzer et al. 2014). With these protocols, a significant portion of patients have a chance to get cured with first-line therapy. For patients not feasible for intensive treatment, dose-adjusted EPOCH-R (rituximab, VP, vincristine, CY, and DOX) may be an option (Roschewski et al. 2020).

88.1.5 Autologous HCT

There are several studies exploring the role of auto HCT in first remission. In a prospective trial, the HOVON group treated 27 patients with two cycles of intensive induction followed by BEAM-conditioned auto HCT for those patients achieving at least a PR (van Imhoff et al. 2005). The 5-year EFS and OS was 73% and 81%, respectively. In a retrospective analysis of 117 patients receiving auto HCT as part of first-line therapy between 1984 and 1994, patients in CR at time of transplant had a 3-year OS of 72% (Sweetenham et al. 1996). In the relapse situation, patients who were chemotherapy sensitive had a 3-year OS of 37% following auto HCT compared to just 7% for those who were chemotherapy resistant (Sweetenham et al. 1996). In summary, auto HCT in BL is feasible, but there is no documented advantage compared to standard combination chemotherapy for patients responding sufficient to first line. Auto HCT may be used to optimize remission in patients with insufficient response or as bridging to allo HCT. In the relapse setting, given the intensive regimens usually used as first line, the difficulty lies in achieving a response good enough to proceed to auto HCT and to collect SC, hence auto HCT is rarely used in BL.

88.1.6 Allogeneic HCT

Indicated in	CR ≥2
Donor	MRS > MUD > mMUD
Conditioning	RIC > MAC
TRM (1 year)	30%
OS (5 years)	30–40%
DFS	35%

Adapted from Peniket et al. (2003)

The kinetics of BL with rapidly progressive disease may not allow for the 4–6 week waiting period during screening, apheresis, and manufacturing necessary for CAR T-cell therapy.

88.2 Lymphoblastic Lymphoma

88.2.1 Definition and Epidemiology

LBL is an aggressive neoplasm of precursor B cells (B-LBL) or T cells (T-LBL) with features of acute leukemia. It accounts for approximately 2% of all NHL. In adults, around 90% of all LBL are T-LBL.

88.2.2 Diagnosis

The diagnosis is based on an LN biopsy. T-LBL is usually TdT positive with variable expression of other T-cell markers (CD7 and CD3 are often positive).

88.2.3 Prognostic Factors

At this time, no convincing prognostic model for these patients is available. Several studies tried to identify risk factors, the following may be associated with an unfavorable outcome: elevated LDH, BM, or CNS involvement, and stage IV disease. There is also an association with lymphotropic viral infections, e.g., HTLV-1. The role of MRD in LBL has not been defined yet, but as we learned from acute leukemias and other lymphomas, persisted MRD positivity might be a predictor of poor outcome.

88.2.4 First-Line Treatment

Standard approaches for patients with LBL are adapted to ALL protocols. These regimens contain multiple drugs, such as corticosteroids, CY, MTX, vincristine, Ara-C, thioguanine, L-asp, VP, nitrosoureas, and anthracyclines. Intrathecal CNS prophylaxis is mandatory. With these protocols, a CR rate more than 80% and a significant chance of cure has been reported [overview in Intermesoli et al. (2022)].

88.2.5 Autologous Transplantation

Since most centers prefer allo HCT over auto HCT because of GVL effect, there are only very few studies evaluating the role of auto HCT in LBL. In CR1, the use of auto HCT as a consolidation may improve relapse-free survival, but has no effect on OS (Sweetenham et al. 2001). In another study in 128 patients with LBL receiving auto HCT, RR at 5 years was 56% (Levine et al. 2003). No documented role in more advanced disease >CR1 is reported either. In conclusion, data for auto HCT in LBL are too scarce to come to firm conclusions.

88.2.6 Allogeneic Transplantation

There is also no established role for allo HCT in patients with LBL. Compared to auto HCT, allo HCT is associated with a higher TRM but lower RR. In 76 patients receiving allo HCT, 5-year RR was 34% (Levine et al. 2003). In this retrospective study, there was no significant difference in OS at 1 year and 5 years between auto HCT and allo HCT. In the relapse setting, bridging with nelarabine might be an option (Candoni et al. 2020). In general, the indication for allo HCT should be based on risk factors, remission, and MRD.

88.3 Peripheral T Cell Lymphomas

88.3.1 Definition and Epidemiology

PTCLs are a very heterogenous group of lymphomas originating from the T cell lineage. They account for approximately 10–15% of all NHL. Because of this low incidence, large randomized studies are difficult to perform.

88.3.2 Diagnosis

The diagnosis, as in any NHL, should be based on a LN biopsy. The differential diagnosis between PTCL and other types of T-NHL is crucial for the outcomes, and in some specific cases, the treatments are very different.

88.3.3 Risk Factors

The IPI is the most commonly used prognostic tool in PTCL. The following factors are associated with worse outcome: age >60 years, ECOG >1, elevated LDH, stages III–IV, and extranodal involvement >1 site. Another prognostic tool is the PIT score, which is calculated by four parameters as age, ECOG, LDH level, and BM involvement. In anaplastic large cell lymphoma (ALCL), the tumors are categorized in ALK+ or ALK− with better prognosis for ALK+ lymphomas.

88.3.4 First-Line Treatment

The primary goal of first-line treatment is to get a deep and continuing remission. Standard regimens are anthracycline-containing combinations like CHOP or CHOEP, achieving a 3-year EFS of 50–70% and a 3-year OS of 75–80% (Schmitz et al. 2010). In CD30-positive PTCL, the addition of brentuximab to chemotherapy improves PFS and OS (Horwitz et al. 2022). In the relapse situation, the overall prognosis of PTLC is dismal, and the optimal treatment for these patients has not been defined yet. Relapse patients not able to receive intensive treatment including HCT may be offered single-agent therapy, e.g., gemcitabine, or in case of CD30 expression brentuximab vedotin.

88.3.5 Autologous Transplantation

Indicated in	CR1 (IPI >1); CR ≥2
Conditioning	BEAC, BEAM, LEAM, CBV
TRM (1 year)	5–7%
REL (3 years)	40–50%
OS (5 years)	70% (CR1) 50% (CR ≥2)
PFS (5 years)	50% (CR1) 20–35% (CR ≥2)

Adapted from Kyriakou et al. (2008), d'Amore et al. (2012), Wilhelm et al. (2016) and Kewalramani et al. (2006)

88.3.6 Allogeneic Transplantation

Indicated in	CR ≥2, relapse post auto HCT
Donor	MRS > MUD > mMUD
Conditioning	RIC > MAC
GVHD prophylaxis	CsA, CsA-MTX, CsA-MMF
Graft failure	<10%
TRM (1 year)	20–25%
REL (3 years)	20–40%
OS (3 years)	40–60%
PFS (3 years)	30–50%

Adapted from Kewalramani et al. (2006), Schmitz et al. (2014, 2018) and Dodero et al. (2012)

88.4 Cutaneous T Cell Lymphomas

88.4.1 Definition and Epidemiology

Cutaneous T cell lymphomas (CTCLs) are a spectrum of lymphoid malignancies characterized by clonal expansion of T-cells primarily involving the skin, which may extend to other organs in advanced forms. CTCLs are relatively rare, with an incidence of approximately six patients per million people (Dobos et al. 2020). The commonest forms (70%) are mycosis fungoides (MF) and Sezary syndrome (SS). Although the exact risk factors for CTCLs remain unclear, an interplay between intrinsic and environmental factors exists.

88.4.2 Diagnosis and Staging

Diagnosis of MF/SS can be challenging and requires clinical data, histopathological findings, and molecular testing. Staging is based on a TNMB classification revised by the ISCL and EORTC (Olsen et al. 2007). Patients with early-stage disease, including skin patches and plaques (stages I–IIA), have an indolent clinical course with excellent long-term survival. Patients with advanced-stage disease, including erythroderma, tumors, or significant blood, nodal or visceral involvement (stages IIB–IV), have poor prognosis with a median survival that does not goes beyond 2–5 years.

88.4.3 Therapeutic Management

The initial management of CTCLs is directed to control symptoms and improve quality of life, particularly in patients with early-stage disease. Some of these patients can have an expectant period, but the vast majority will start with skin-directed therapies, such as topical steroids, total skin electron beam therapy, ultraviolet B phototherapy, PUVA, and others (Quaglino et al. 2021). Patients with refractory early-stage or advanced-stage disease have multiple therapeutic options, albeit most with low response rates and duration (Giordano and Pagano 2022). Some novel targeted therapies such as brentuximab vedotin and mogamulizumab have emerged with promising results. However, allogeneic HCT remains the only potential curative option.

88.4.4 Allogeneic HCT

EBMT has led international efforts to examine the role of allogeneic HCT in patients with CTCLs (Duarte et al. 2008, 2010, 2014; Domingo-Domenech et al. 2021). Overall, high relapse rate remains the major cause of failure and needs to be improved with better strategies before and after transplant. Nevertheless, with a long follow-up in survivors of more than 6 years, we have shown that allogeneic HCT can rescue over one third of patients with advanced-stage and refractory CTCLs, and it remains a clinical option for patients with advanced-stage CTCLs (Snowden et al. 2022). Also, over time there is a trend to offer allogeneic HCT earlier in the course of the disease, including older patients as well as the use of more unrelated donors.

88.5 HIV-Associated Lymphomas

88.5.1 Definition and Epidemiology

Patients infected with HIV have an increased risk of developing NHLs as compared to the general population. The most frequent subtypes are DLBCL and BL which both are AIDS-defining

illnesses, and HL, which is one of the non-AIDS-defining malignancies.

88.5.2 Risk Factors

Factors that determine prognosis in patients with HIV-related lymphoma are the same as in the general population. Additionally, a low CD4+ T-cell count and an uncontrolled HIV viral load are independent risk factors for HIV-lymphomas.

88.5.3 First-Line Therapy

The availability of combination antiretroviral therapy (cART), along with the better management of opportunistic infections, allows HIV-infected patients with lymphoma to receive the same treatment approaches as HIV-uninfected patients. The consequent use of cART during therapy is of major important for successful treatment. The indication for HCT has to be discussed in the relapse situation.

88.5.4 Autologous Transplantation

Indicated in	CR ≥2; same indications as in general population with the same type of lymphoma
Conditioning	BEAC, BEAM, LEAM, CBV
TRM (1 year)	5–7%
REL (3 years)	30–40%
OS (5 years)	50–60%
PFS (5 years)	50–60%

Adapted from Hübel et al. (2019), Diez-Martin et al. (2009) and Balsalobre et al. (2009)

88.5.5 Allogeneic Transplantation

Experience on the use of allo HCT in patients with lymphoma and HIV infection is limited and no definitive recommendation can be given at this time. There are some case reports or small retrospective analysis showing that allo HCT in HIV positive patients using MRD, MUD, or cord blood is feasible, but application of cART and viroimmunological reconstitution is a matter of debate. In a report of five HIV-positive patients who underwent allo HCT with various hematologic malignancies, there was no TRM or major infections (Mulanovich et al. 2016). HIV virus load remained undetectable with continuous cART. Three patients relapsed 6, 7, and 13 months after transplant, and two were alive and well after 42 and 55 months.

There are a few case reports available showing that the application of CAR-T-cell therapy is feasible in HIV-related lymphomas.

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